

DISP: Testing Report

1. Introduction

What is being tested

The completed operational BPMN process model for the ProBuilds tool hiring system is being tested, including all user task forms, service tasks, message events, and gateways. This covers the end-to-end flow of the system.

Scope of testing

- All user task forms and their input validation
- All service tasks and their outputs
- Gateway conditions and process routing (correct paths taken based on user input)
- Message events and their triggers
- Database interactions

2. Testing Strategy

Manual testing was selected as the primary testing strategy for the ProBuilds system. According to Thant and Tin (2023), manual testing involves testers executing test cases manually, without the use of software tools, to verify the functionality of the software.

This approach was chosen for several reasons. First, the ProBuilds system is driven by user task forms in Camunda Tasklist which require a human to fill in and submit, making automated testing impractical for this project. Second, Thant and Tin (2023) note that manual testing allows

testers to explore the software more thoroughly, and it is often more effective in detecting defects and errors that may be missed by automatic testing. Third, manual testing is more flexible than automated testing (Thant and Tin, 2023), which was important as the system was still being refined during testing. Additionally, evidence was collected at every stage through screenshots of Camunda Operate and Tasklist, providing clear and detailed documentation of the testing process.

Manual Testing vs Automated Testing Comparison Table

	Manual Testing	Automated Testing
Human Input	Required, suits user task forms	Cannot fill in forms automatically
Flexibility	Adaptability to system changes	Limited to pre-written scripts
Cost	Cost effective, no extra tools needed	Requires specialised testing tools
Defect detection	More effective at finding complex issues	May miss issues requiring human judgement
Speed	Slower than automated	Faster for repetitive tasks
Suitable for this project	Yes	No

Based on the comparison above, manual testing was selected as the most appropriate strategy for this project (Thant and Tin, 2023).

3. Functional Requirements

FR ID	Functional Requirement Description
FR01	The customer shall be able to select in-store or online.
FR02	The system shall allow the customer to choose whether they want to hire or purchase a tool.
FR03	The system shall retrieve the available tools and its details from the database based on customer selection (hire or purchase).
FR04	If the customer chooses to purchase, the system shall limit the quantity the user can select based on available stock.
FR05	If the customer chooses to hire, the system shall allow the customer to hire only one tool.
FR06	Existing members shall be able to enter their member ID before making a payment.
FR07	The system shall allow non-members to choose whether they want to register.
FR08	Non-members who want to register shall be able to fill out a registration form.
FR09	Non-members who don't want to register shall be able to enter the required details to proceed with their order.
FR10	The system shall validate the membership number to confirm that the user is part of the membership.
FR11	The system shall register and save new member details in its database.
FR12	For normal customers(non-members) who are purchasing, the system shall calculate the full price with no discounts, including a delivery fee if needed.
FR13	For existing members, the system shall calculate the price applying any eligible discount based on their loyalty points.
FR14	For new members, the system shall calculate the price, including the £10 membership registration fee.
FR15	For customers who are hiring, the system shall calculate the total price based on the hire days.
FR16	Customers shall be able to choose between cash and card payment.
FR17	If it's a card payment, the system shall get the customer card details and validate them.

FR18	After entering card details, and the customer is not eligible for financial support or doesn't choose to apply for financial support, the system shall process the payment by checking the card expiry date.
FR19	For existing or new members with purchases up to £100, the system shall allow the customer to apply for financial support, choosing the instalment plan for 6 or 12 months.
FR20	If the member chooses to apply for financial support, the system shall create the credit application, including the application ID and send it to the FinTrust company.
FR21	The system shall receive the application and check whether the customer already has an active credit plan.
FR22	If the customer doesn't have an active credit plan, the system shall approve the credit application and set up a credit plan.
FR23	If the customer has an active credit plan, the system shall deny the credit application, and it shall route denied credit applications back to normal payment processing and the customer cannot apply again for financial support.
FR24	After payment is completed, the system shall create and save the order to the database.
FR25	If the order is a purchase and the customer wants the items to be delivered, the logistics team is responsible for delivering the order and recording it in the system when it's delivered.
FR26	If the order is a purchase and the customer chooses to pick up their order, the retail store shall notify the customer when it's ready.
FR27	If the order is a hiring, the hiring division shall prepare the tools by issuing the tools with a return date and a condition log.
FR28	When the hiring order is collected by the customer, the system shall activate the rental agreement.
FR29	When the customer returns the tools, the system shall allow the employee to tag and log the tools for maintenance.
FR30	FixPro collects tools every Tuesday, and after inspection, the system shall allow the technicians to categorise the tools based on the service they need and add any additional information, such as
FR31	When a tool needs to be repaired, and it exceeds £50, the system shall allow FixPro technicians to prepare and submit an authorisation request by entering the request date, subject/title, description, urgency level, and estimated repair cost.

FR32	The system shall allow ProBuild's tool hire manager to review an authorisation request and select whether to approve or decline it, and the decision shall be sent back to FixPro.
FR33	The system shall allow FixPro staff to create and submit a service report containing tool serial number, service details, and share it with ProBuild.
FR34	The system shall update the tool instance status based on the maintenance option selected. If tools are decommissioned, the status shall be set to retired, and for other tools, the status shall be set to available.

4. Test Cases Table

<u>Test ID</u>	<u>Task Name</u>	<u>Task Type</u>	<u>Description</u>	<u>Input/Condition</u>	<u>Expected results</u>	<u>Actual results</u>	<u>Status</u>
<u>TC01</u>	Decide in-store or online	User task	The customer selects in-store or online as a shopping option, and hire or sale as an order type, so they can decide where to shop, and what to shop for.	Customer selects “in-store” or “online” Customer selects sale	In_store_online=”in_store” or “online” stored as process variable, and toolType is stored as “SALE”, process route to check_availability service task	As expected	Pass
<u>TC02</u>	Decide in-store or online	User task	The customer selects in-store or online as a shopping option, and hire or sale as an order type, so they can decide where to shop, and what to shop for.	Customer selects “in-store” or “online” Customer selects hire	In_store_online= “in_store” or ”online” stored as process variable, and toolType is stored as “HIRE”, process route to check_availability service task	As expected	Pass
<u>TC03</u>	Check availability	Service task	Fetches available tools from DB based on tool type, so the customer	Customer selects toolType=”HIRE” or “SALE”	Returns list of available tools with label, value, price, and limiting quantity user can select based off available stock or only 1 if toolType is “HIRE”.	As expected	Pass

			can select the tool they want.				
<u>TC04</u>	Check availability	User task	Customer selects a tool based off tool type and available tools as well as quantity of tool.	Customer selects a tool from the dropdown, if purchase customer can select quantity but limited available stock.	Customer can select tool from dropdown, quantity limited to 1 if hire, multiple quantity allowed for purchase, process continues to get customer information.	As expected	Pass
<u>TC05</u>	Get customer information/membership number	User Task	Existing member enters valid membership number so they can access their membership account	Customer selects that they are a member Customer enters membership number	Membership is validated and discount is applied later If membership is not validated the user fills out the form again	As expected	Pass
<u>TC06</u>	Get customer information/membership number	User Task	Customer is not a member and wants to register	Customer selects that they aren't a member Customer chooses to register Customer inputs name,	Process routes to opt for membership gateway, then continues to register_customer service task	As expected	Pass

				surname, phone number, and email address			
<u>TC07</u>	Get customer information/membership number	User Task	Customer is not a member and doesn't want to register	Customer selects that they aren't a member and chooses not to register, fills in name, email, phone (no DOB required)	Basic customer details stored as process variables, process routes to calculate price	As expected	Pass
<u>TC08</u>	Validate membership	Service Task	Validate Membership Number against database to ensure that customer is a member	Membership number passed from get_customer_information form	Returns true, process routes to calculate_price via membership gateway If Customer doesn't want to be registered, loop back If user is not a member, skip	As expected	Pass
<u>TC09</u>	Validate membership	Service Task	Membershipnumber not found	Membership number not found in MemberRepository	Returns false, process routes to get Customer Information/Membership Number Task	As expected	Pass
<u>TC10</u>	Register customer	Service task	Registers a membership for a customer	All valid details passed from registration form (name, DOB, email, phone)	New member is saved to the DB with generated Membership Number and process continues	As expected	Pass

<u>TC11</u>	Calculate price	Service task	Calculates full price for normal customer hire with pickup	ToolName=valid tool, quantity=1, membershipNumber=null, distributionType="pickup", hireDays=1	TotalPrice=ToolPrice * HireDays, discountApplied=0, membershipFee=0, deliveryFee=0	As expected	Pass
<u>TC11.1</u>	Calculate price	Service task	Calculates price for sale with delivery fee	ToolName=valid tool, quantity=1, distributionType="delivery"	deliveryFee= £4, TotalPrice=ToolPrice * Quantity + deliveryFee,	As expected	Pass
<u>TC12</u>	Calculate price	Service task	Calculates sale price for existing member with no loyalty points	Member has 0 points, buys Power Drill x1	totalPrice= £45, discountApplied=0, discountAmount=0	As expected	Pass
<u>TC13</u>	Calculate price	Service task	Calculates sale price for a newly registered member with £10 membership fee	Register_member=true, buys Power Drill x1, 0 loyalty points	totalPrice= £55 (£45 + £10 membership fee), membershipFee=10, discountApplied=0	As expected	Pass
<u>TC14</u>	Calculate price	Service task	Calculates sale price for member with loyalty points, gets a discount	Member has 100 loyalty points, buys Power Drill x1 (£1 = 1 point, 100 points = 1% discount)	FinalPrice= £45 rounded, discountApplied=1, pointsUsed=100, discountAmount= £0 rounded	As expected	Pass

<u>TC15</u>	Get payment details	User task	Customer selects card as payment method but enter invalid or missing card details	paymentMethod = "card", cardNumber is not 16 digits, CVV is less than 3 digits, customerName is empty	Form validation error shown, form cannot be submitted	As expected	Pass
<u>TC16</u>	Get payment details	User task	Customer selects card as payment method with order over £100, ticks "I need financial support", and selects Installment Plan	OrderTotal > £100, paymentMethod="card", customer ticks "I need financial support" (needFinance=true) Installment_Plan = 6 Months	Financial support checkbox visible, instalment plan dropdown appears with 6 months and 12 months option	As expected	Pass
<u>TC17</u>	Get payment details	User task	Customer selects cash as payment method or has order total under £100 and does not tick "I need financial support"	OrderTotal < £100 or paymentMethod="cash", customer does not tick "I need financial support" (needFinance=false)	Financial support checkbox hidden, instalment plan hidden, card fields hidden if cash selected	As expected	Pass

<u>TC18</u>	Get payment details	User task	Customer whose credit application was denied re-enters payment details	Decision=denied, customer routed back to get_payment_details	Financial support option no longer available, customer can only pay by card or cash	As expected	Pass
<u>TC19</u>	Process credit application	Service task	After customer fills in payment details and ticks "I need financial support", generates a unique application ID for credit application	Customer ticks needFinance=true, valid card payment	ApplicationID generated via UUID, all process variables copied, process routes to Send Credit Application Message Event	As expected	Pass
<u>TC20</u>	Process customer's credit application	Service task	Checks if customer has an existing active credit plan – none found	Customer has no existing plan	Returns false, process routes to generate credit plan	As expected	Pass
<u>TC21</u>	Process customer's credit application	Service task	Checks if customer has an existing active credit plan – existing plan found	Customer already has an active credit plan	Returns true, credit application denied, decision sent back to customer	As expected	Pass

<u>TC22</u>	Set up credit plan	Service task	Sets up a new credit plan for an approved credit application	Customer has no existing credit plan, instalment plan selected (6 or 12 months)	Credit plan saved to DB, monthly payment calculated, decision sent back to customer	As expected	Pass
<u>TC23</u>	Process payment	Service task	Card payment with valid details and non-expired card	PaymentMethod="card", all fields filled, card not expired, totalPrice>0	PaymentCompleted=true, amountPaid=totalPrice, process routes to payment complete gateway	As expected	Pass
<u>TC24</u>	Process payment	Service task	Card payment with expired card	PaymentMethod="card", all fields filled but card is expired	PaymentCompleted=false, process loops back to get payment details	As expected	Pass
<u>TC25</u>	Process payment	Service task	Cash payment with valid total	PaymentMethod="cash", totalPrice>0	PaymentCompleted=true, amountPaid=totalPrice, process routes to payment complete gateway	As expected	Pass
<u>TC26</u>	Create order	Service task	Creates a hire order after payment is completed	ToolType="HIRE", valid toolName, paymentMethod, distributionType, totalPrice, quantity	Hire order saved to DB, unique orderId generated, status=PENDING, ToolInstanceStatus=HIRED, order details sent to ProBuild, process routes to hire division	As expected	Pass
<u>TC27</u>	Create order	Service task	Creates a sale order after payment is completed with delivery	ToolType="SALE", distributionType="delivery", valid toolName, paymentMethod,	Sale order saved to DB, unique orderId generated, status=PENDING, ToolInstanceStatus= SOLD, order	As expected	Pass

				distributionType, totalPrice, quantity	details sent to ProBuild, process routes to logistics team for delivery		
<u>TC28</u>	Create order	Service task	Creates a sale order after payment is completed with pickup	ToolType="SALE", distributionType="pickup", valid toolName, paymentMethod, totalPrice, quantity	Sale order saved to DB, unique orderId generated, status=PENDING, ToolInstanceStatus=SOLD, order details sent to ProBuild, customer notified when ready for pickup	As expected	Pass
<u>TC29</u>	Mark item as delivered	User task	Logistics team marks an item as delivered to the customer after successful delivery	Customer information is displayed and status checkbox is ticked by the logistics staff which means it's delivered	Order status updated to DELIVERED, process completes for the delivery path	As expected	Pass
<u>TC30</u>	Issue tools with return date & condition log	User task	Staff from tool hire division issues hired tools to customer by recording tool condition and setting expected return date	Tool and borrower are pre- filled, staff fills in condition at issue, condition notes, and expected return date	Information will get saved in DB, hire order updated, process proceeds to notify customer that tools are ready for collection	As expected	Pass
<u>TC31</u>	Activate rental agreement	Service task	After customer picks up the hired	OrderId passed from process variables,	Hire order status is changed from PENDING to PICKED_UP,	As expected	Pass

			tools, the system activates the rental agreement and updates the hire order status	conditionAtIssue, notesOnIssue, IssueDate and expectedReturnDate passed from previous user task	conditionAtIssue, conditionNotes, issueDate and expectedReturnDate saved to DB		
<u>TC32</u>	Tag and log tools for maintenance	User task	the tool hire division staff prepares digital tag and logs returned tools for maintenance	Serial Number pre-filled, staff optionally fills in condition notes, fault description and expected service type	tools logged for maintenance, ToolInstanceStatus changes from HIRED to MAINTENANCE, process continues to store tools in Service Bay	As expected	Pass
<u>TC33</u>	Categorise service	User task	FixPro technician categorises the service needed for the returned tools after inspection	Tool serial numbers pre-filled, FixPro technician has inspected the tools and selects the service classification which are repairNeeded, routineMaintenance, or decommission	Process routes to the path based on service classification selected, maintenanceOption stored as process variable	As expected	Pass
<u>TC34</u>	Prepare authorisation request	User task	FixPro technician fills in the form for request details	The FixPro technician inputs date of request, subject/title of the request, description, urgency level	Form is filled and is sent to the ProBuild	As expected	Pass

				(low, medium, high), repair cost			
<u>TC35</u>	Approve or decline authorization	User task	ProBuild's tool hire manager makes the decision to approve or decline the authorisation request	Authorisation request received from FixPro, tool hire manager reviews the request details and selects approve or decline	The decision of the request is sent to FixPro and they proceed the process based on the decision made	As expected	Pass
<u>TC36</u>	Create and share service reports	User task	After finishing the servicing of the tool, the staff creates and shares the service reports to ProBuild	Tool serial number, service type, service description, and total cost for service pre-filled. Staff inputs service date and total service hours.	Report filled and saved, tools and report are sent to ProBuild, process continues	As expected	Pass
<u>TC37</u>	Mark tools as available for hire in online system	Service task	After the tools are returned and report is shared, tool instances are marked as available or retired based on maintenance option	OrderId passed from process variables, maintenanceOption from categorise service form	If maintenanceOption="decommissioned" --> ToolInstanceStatus updated to RETIRED. Otherwise --> ToolInstanceStatus updated to AVAILABLE	As expected	Pass

5. Traceability Matrix

<u>FR ID</u>	<u>Test Case ID(s)</u>	<u>BPMN Task / Event Name</u>	<u>Status</u>
FR01	TC01, TC02	Decide In-Store or Online	Pass
FR02	TC01, TC02	Decide In-Store or Online	Pass
FR03	TC03, TC04	Check Availability	Pass
FR04	TC03, TC04	Check Availability (Purchase Qty Limit)	Pass
FR05	TC03, TC04	Check Availability (Hire – One Tool)	Pass
FR06	TC05	Get Customer Information / Membership Number	Pass
FR07	TC06, TC07	Get Customer Information / Membership Number	Pass
FR08	TC06	Get Customer Information / Registration Form	Pass
FR09	TC07	Get Customer Information / No Registration	Pass
FR10	TC08, TC09	Validate Membership	Pass
FR11	TC10	Register Customer	Pass

FR12	TC11, TC11.1	Calculate Price (Non-Member / Full Price)	Pass
FR13	TC12, TC14	Calculate Price (Existing Member / Loyalty Discount)	Pass
FR14	TC13	Calculate Price (New Member / £10 Membership Fee)	Pass
FR15	TC11	Calculate Price (Hire – Based on Hire Days)	Pass
FR16	TC15, TC17, TC25	Get Payment Details (Cash / Card Selection)	Pass
FR17	TC15, TC16	Get Payment Details (Card Validation)	Pass
FR18	TC23, TC24	Process Payment (Card – No Financial Support)	Pass
FR19	TC16, TC17, TC18	Get Payment Details (Financial Support Eligibility)	Pass
FR20	TC19	Process Credit Application / Send Credit Application Message Event	Pass
FR21	TC20, TC21	Process Customer's Credit Application (Active Plan Check)	Pass
FR22	TC22	Set Up Credit Plan (Approved)	Pass

FR23	TC21, TC18	Process Customer's Credit Application (Denied) / Get Payment Details	Pass
FR24	TC26, TC27, TC28	Create Order	Pass
FR25	TC29	Mark Item as Delivered	Pass
FR26	TC28	Create Order (Pickup Notification)	Pass
FR27	TC30	Issue Tools with Return Date & Condition Log	Pass
FR28	TC31	Activate Rental Agreement	Pass
FR29	TC32	Tag and Log Tools for Maintenance	Pass
FR30	TC33	Categorise Service	Pass
FR31	TC34	Prepare Authorisation Request	Pass
FR32	TC35	Approve or Decline Authorisation	Pass
FR33	TC36	Create and Share Service Reports	Pass
FR34	TC37	Mark Tools as Available / Retired in Online System	Pass

6. Test Results

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?

Processes

Browse and run processes published by your organization.

Search processes

All Processes

tool hire services

Process_k9l4ulf

Start process

MainProcess

Start process

Process_f7nbpgk

Start process

Process_w2dmrrp_new

Start process

[Load more](#)

Operate: Process Instances

localhost

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Tasklist

Tasks

Processes

Non-production license

Non-commercial license

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Decide in-store or online

MainProcess

Assigned to me

Unassign

Task

Process

History

Select shopping option*

in-store

Hire or Sale*

Select

Hire

Sale

Auto-select first available task

Complete Task

Details

Creation date

07 May 2026 - 4:19 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?👤

All open tasks⇅

Check availability

MainProcess

Me

Now

Medium

TaskProcessHistory

Hire or Sale

HIRE

Tool Name*

Select

Cement Mixer

Scaffolding

Pressure Washer

Tile Cutter

Heavy-Duty Drill

*£2 delivery fee

Days to Hire for?*

-

+

Auto-select first available task

Complete Task

Details

Creation date

07 May 2026 - 4:22 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?

All open tasks

Check availability

MainProcess

Me

Now

Medium

Task

Process

History

Hire or Sale

HIRE

Tool Name*

Scaffolding

Quantity*

2

Field must have maximum value of 1.

Distribution Type*

Pickup

Pickup

Days to Hire for?*

2

Auto-select first available task

Complete Task

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Check availability

MainProcess

Assigned to me

Unassign

Details

Creation date

07 May 2026 - 4:22 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Tasklist

Tasks

Processes

Non-production license

Non-commercial license

?

All open tasks

get customer information/
membership number

MainProcess

Me

Now

Medium

Task

Process

History

Customer Information

Are you a member*

Yes

Membership Number*

100023

Auto-select first available task

Complete Task

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

get customer information/ membership number

MainProcess

Assigned to me

Unassign

Details

Creation date

07 May 2026 - 4:23 AM

Candidates

No candidates

Priority

Medium

Due date

No due date



MainProcess

Me

📅 1 minute ago

Medium



Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

×

get customer information/ membership number

MainProcess

 Assigned to me

Task

Process

History

Customer Information

Are you a member*

No

×

- ✓ Register to become a member (£10 per year)

First Name*

Amin

Last Name*

Bashir

Date of Birth*

10/02/199

1

Email Address*

aminbashir@yahoo.mail

Phone Number

79288273636

©

☐

Complete Task

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?

All open tasks

get payment details

MainProcess

Me

Medium

1 minute ago

TaskProcessHistory

£ 130

Payment Method*

Credit Card

Cardholder Name*

Amin Bashir

Card Number*

1234875454892382

Expiry Date (mm/yy)*

02/29

CVV*

234

☒ I confirm the payment details are correct.*

I am liable for the borrowed tools; any damage incurred in the rental period will be attributed to me.

☒ I agree*

Details

Creation date

07 May 2026 - 4:25 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Auto-select first available task

Tasklist

Tasks

Processes

Non-production license

Non-commercial license

?

All open tasks

get payment details

MainProcess

Me

Medium

1 minute ago

Task

Process

History

Complete Payment

Total Price

£ 130

Payment Method*

Cash

I am liable for the borrowed tools; any damage incurred in the rental period will be attributed to me.

☒ I agree*

Complete Task

Details

Creation date

07 May 2026 - 4:25 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

×

get payment details

MainProcess

Assigned to me

Unassign

Auto-select first available task



MainProcess

Process Instance Key
2251799813831151

Version
29

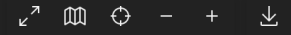
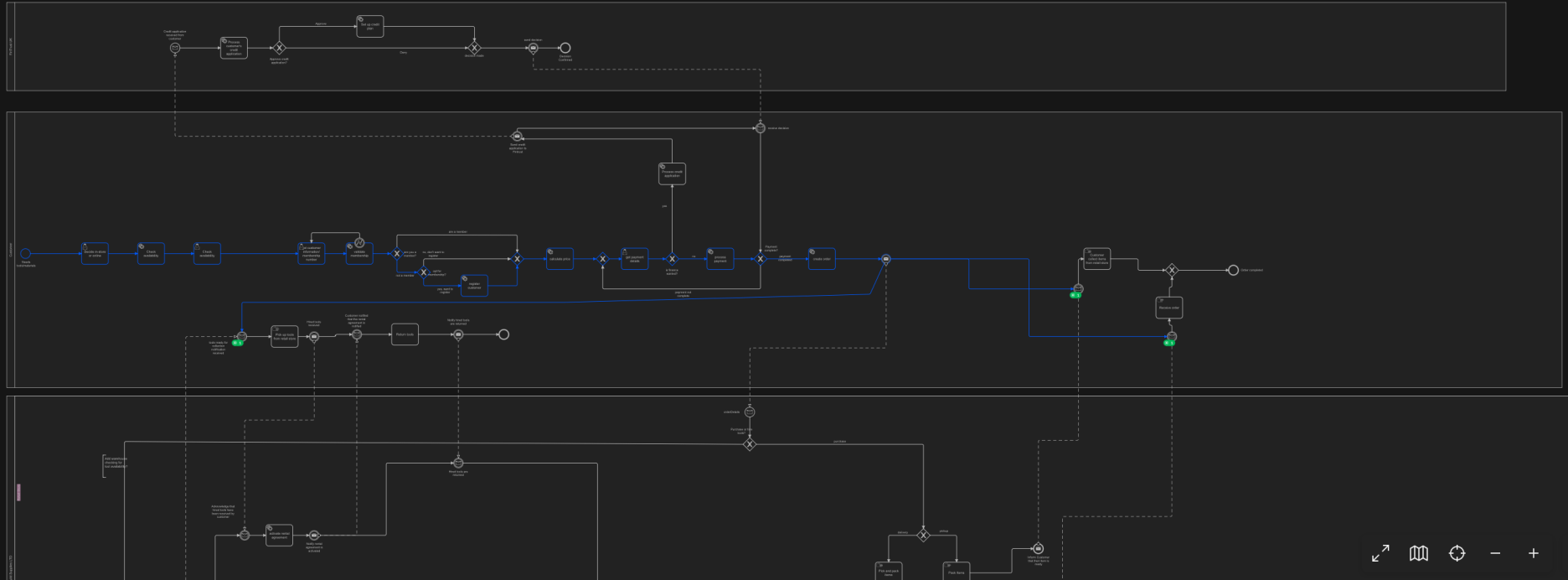
Start Date
2026-05-07 04:19:31

Called Instances
None

Cancel

Modify

Migrate



Tasklist

Tasks

Processes

Non-production license

Non-commercial license

?

All open tasks

Issue tools with return date & condition log

tool hire services

Me

Medium

1 minute ago

Task

Process

History

Auto-select first available task

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

×

Issue tools with return date & condition log

tool hire services

Assigned to me

Unassign

Task

Process

History

Tool Checkout & Return Form

Tool Information

Order ID

73aefbd5-df33-4aef-9285-dc03d65e80ca

Tool Name*

Scaffolding

Borrower Information

Name*

Amin

Surname*

Bashir

Quantity

1

Issue Log

Issue Date*

Details

Creation date

07 May 2026 - 4:27 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Tasklist

Tasks

Processes

Non-production license

Non-commercial license

?

All open tasks

Issue tools with return date & condition log

tool hire services

Me

Medium

1 minute ago

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

×

Issue tools with return date & condition log

tool hire services

Assigned to me

Unassign

Task

Process

History

Issue Date*

07/05/2026

Condition at Issue*

New

×

▼

Notes on Condition at Issue

No notes

Hire Days

2

Expected Return Date*

09/05/2026

Complete Task

Details

Creation date

07 May 2026 - 4:27 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Auto-select first available task

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?

All open tasks

Tag and log tools for maintenance
tool hire services

Me

Now

Medium

Auto-select first available task

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Tag and log tools for maintenance

tool hire services

Assigned to meUnassign

TaskProcessHistory

Digital Handover Form

Tool Serial Number

SC-11-007

Condition Notes

Scratches on the side

Fault Description

Minor

Select expected maintenance option*

Routine Maintenance

Complete Task

Details

Creation date

07 May 2026 - 4:29 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?

All open tasks

Categoryse service

Process_f7nbpgk

Me

Now

Medium

Task

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Auto-select first available task

Complete Task

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Categoryse service

Process_f7nbpgk

Assigned to meUnassign

Maintenance Issue Form

Tool Serial Number

SC-11-007

Select maintenance option*

Select

Routine Maintenance

Decommissioned

Repair Needed

Details

Creation date

07 May 2026 - 4:31 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

All open tasks



Categorise service

Process_f7nbpkg

Me

Medium

📅 1 minute ago

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.



Categorise service

Process_f7nbpkg

 Assigned to me

Task

Process

History

Maintenance Issue Form

Tool Serial Number

SC-11-007

Select maintenance option*

Repair Needed



Description

Nothing to say

Total Cost for Service*

£ 200

☐ Auto-select first available task

Complete Task

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?👤

All open tasks

Prepare authorisation request

Process_f7nbpjk

Me

1 minute ago

Medium

TaskProcessHistory

Request Details

Date of Request*

07/05/2026

Subject/Title of Request*

Request for outer cover to be changed

Detailed Description and Business Justification*

The longer it goes without repair, the more damage it will cause. This is insurance for your tool.

Urgency Level*

Low

Low

Medium

High

Auto-select first available task

🔔 Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Prepare authorisation request

Process_f7nbpjk

Assigned to me

Unassign

Details

Creation date

07 May 2026 - 4:32 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Tasklist

Tasks

Processes

Non-production license

Non-commercial license

?

All open tasks

Approve or Decline Authorization

tool hire services

Me

Medium

1 minute ago

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

×

Approve or Decline Authorization

tool hire services

Assigned to me

Unassign

Task

Process

History

Request Details

Date of Request*

07/05/2026

Subject/Title of Request*

Request for outer cover to be changed

Urgency Level*

low

Approval Information

Approver's Name

Martin Ø

Approver's Title

Tool Hire Manager

Approved?

Details

Creation date

07 May 2026 - 4:34 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Auto-select first available task

Tasklist

TasksProcesses

Non-production licenseNon-commercial license?

All open tasks

Approve or Decline Authorization

tool hire services

Me

2 minutes ago

Medium

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Approve or Decline Authorization

tool hire services

Assigned to meUnassign

TaskProcessHistory

Approver's Name

Martin Ø

Approver's Title

Tool Hire Manager

Approved?

Date of Decision

07/05/2026

Approver's Comments

200 for some scratches? Let it be. I will have a word with whoever is in charge over there and I mean it.

Attachments

Please attach any supporting documents (e.g., quotes, invoices, forms) to the process instance after submitting this form.

File picker

Browse

No files selected

Auto-select first available task

Complete Task

Details

Creation date

07 May 2026 - 4:34 AM

Candidates

No candidates

Priority

Medium

Due date

No due date

Operate: Process Instances

localhost

DISP/README.md at main · ZainabMB/DISP

Tasklist

Tasks

Processes

Non-production license

Non-commercial license

All open tasks

create and share service reports

Process_f7nbpgk

Me

Now

Medium

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

create and share service reports

Process_f7nbpgk

Assigned to me

Unassign

Task

Process

History

Service Summary Report

Tool Serial Number

SC-11-007

Service Type

repairNeeded

Service Description

Nothing to say

Service Date

07/05/2026

Total Service Hours

0

-

+

Total Cost for Service

£ 200

-

+

Details

Creation date

07 May 2026 - 4:37 AM

Candidates

No candidates

Priority

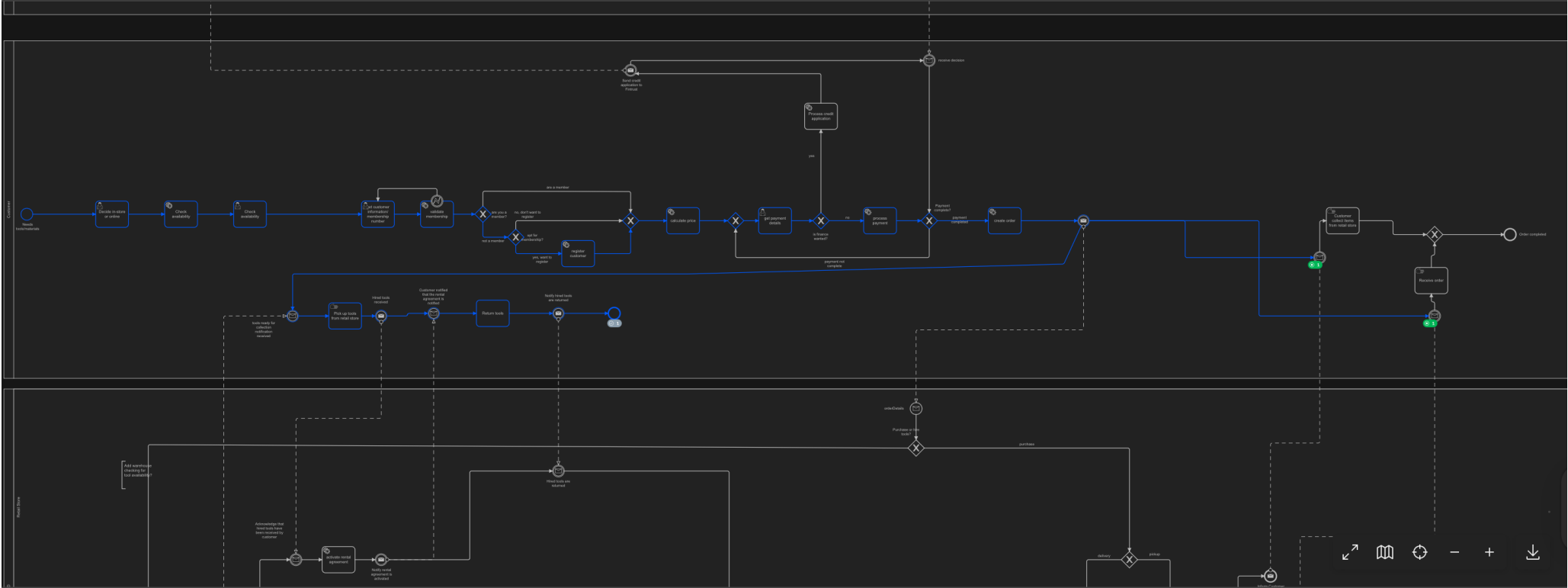
Medium

Due date

No due date

Auto-select first available task

This should show 0 because nothing was paid because it was not approved



Instance History		Variables		Listeners	Operations Log
MainProcess		Name		Value	
Needs tools/materials		amountPaid		130.0	
Decide in-store or online		availableTools		[{"quantity":4,"price":45.0,"label":"Cement"}]	
Check availability		Add Variable +		Copy variables	
Check availability					

✔ Process_f7nbpkg	Process Instance Key	Version	Start Date	End Date	Called Instances
	2251799813834298	29	2026-05-07 04:31:02	2026-05-07 04:40:00	None

Instance History

☒ End date ☒ Execution count

Variables Listeners Operations Log

Name

Value

☒ End date ☒ Execution count

- ✓  Tools are returned
- ✓  re-scan tools to inventory system
- ✓  mark tools as available for hire in online system
- ✓  Tool hire process completed

Listeners Operations Log

7. Defects and Limitations

Defects

- The main process instance does not terminate cleanly upon completion, as active tokens remain at the end of the process flow. This is likely caused by unconsumed tokens across parallel paths involving multiple pools and lanes, causing process instances to appear as still active in Camunda Operate rather than completed.

Limitations

- Due to constraints of the Camunda runtime environment, it was not possible to fully implement a rental agreement that dynamically tracks return dates or automates follow-up actions such as overdue flagging or reminders.
- The system only supports the selection of one tool type per transaction. Customers requiring a mix of hire and purchase items must initiate separate process instances.
- Customers are restricted to hiring one tool per transaction, which is a practical limitation for those requiring multiple tools. Expanding this would require significant changes to the hire, tracking, and return logic.
- Loyalty points are used in price calculation but are not deducted or accrued in the database after payment is completed. A full loyalty point lifecycle, which involves earning, spending, and balance updates, was outside the scope of this iteration.
- The maintenance workflow handles one tool per process instance. As FixPro collects tools in batches, multiple instances must be run individually rather than processing tools as a group.
- No dedicated database schema exists for FixPro. In a real-world integration, FixPro would operate its own independent data store, which this implementation does not reflect.
- Rather than using separate databases for each organisational pool (ProBuilds, FixPro, FinTrust, bank), a single database with separate schemas was used. This was a deliberate decision to preserve codebase integrity, though it would not be suitable for a fully deployed multi-organisation system.
- The system does not support annual membership renewal or the associated £10 renewal fee. Implementing this would require timer-based events or a scheduled process outside the current BPMN model.

- Differences in operating systems (macOS vs. Windows) and database software (PostgreSQL vs. MySQL) among team members occasionally caused compatibility issues with file formatting, environment configuration, and query syntax during development.

Reference List

Thant, K.S. and Tin, K. (2023). THE IMPACT OF MANUAL AND AUTOMATIC TESTING ON SOFTWARE TESTING EFFICIENCY AND EFFECTIVENESS. [online] 3(3), pp.88–93. Available at:

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